

Chapter 02: Atomic Structure & Interatomic Bonding

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Materials Science (MIAE 221)

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapter 2

1. Executive Overview & First-Principles Philosophy

Why do solid objects have defined shapes and resist being crushed into zero volume or pulled apart into a cloud of atoms? Why does diamond scratch every other substance while lead can be scratched with a fingernail? Why does tungsten melt at 3410°C while mercury melts at -39°C ?

All macroscopic mechanical, thermal, electrical, and chemical properties of engineering materials are direct manifestations of **atomic structure and the interatomic forces holding atoms together**:

- **Interatomic forces** govern stiffness (Young's modulus E), thermal expansion (α_l), and melting temperature (T_m).
- **Electronic structures and bond directionality** dictate electrical conductivity (σ), optical transparency, and whether a material deforms plastically with high ductility (like copper) or fractures catastrophically with zero ductility (like quartz glass).

At the atomic scale, matter operates under a continuous dynamic balance between **attractive forces** (pulling oppositely charged electrons and nuclei together) and **repulsive forces** (quantum mechanical Pauli exclusion resisting the overlap of filled electron clouds). Solid materials settle at an equilibrium separation distance (r_0) where the net force is identically zero and the net potential energy is at an absolute minimum.

2. Mathematical Framework & Atomic Mechanics (Callister §2.2)

2.1 The Architecture of the Atom

A. Fundamental Subatomic Quantities

An atom consists of a dense, positively charged nucleus surrounded by a cloud of negatively charged electrons:

- Protons (p^+): Charge = $+1.602 \times 10^{-19} \text{ C}$, mass = $1.673 \times 10^{-27} \text{ kg}$.
- Neutrons (n^0): Charge = 0, mass = $1.675 \times 10^{-27} \text{ kg}$.
- Electrons (e^-): Charge = $-1.602 \times 10^{-19} \text{ C}$, mass = $9.109 \times 10^{-31} \text{ kg}$ (roughly $1/1836$ the mass of a proton).
- **Atomic Number (Z)**: The number of protons in the nucleus, defining the chemical identity of the element.
- **Atomic Mass (A)**: The sum of protons and neutrons: $A \cong Z + N$.

- **The Mole & Avogadro's Number (N_A):** One mole of any substance contains:

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

The atomic weight of an element in grams per mole (g/mol) equals the mass in atomic mass units (amu) of a single atom ($1 \text{ amu} = 1.6605 \times 10^{-24} \text{ g}$).

B. Wave-Mechanical Model & Quantum Numbers

In classical Bohr theory, electrons were envisioned as orbiting the nucleus in fixed circular planetary tracks. Modern quantum mechanics, established by de Broglie and Schrödinger, treats electrons with **wave-particle duality**: an electron's position cannot be localized precisely (Heisenberg Uncertainty Principle), but is described by a spatial probability density distribution called an **atomic orbital**.

Every electron in an atom is completely specified by a set of **four quantum numbers**:

Quantum Number	Symbol	Allowed Values	Physical Meaning
Principal	n	$1, 2, 3, 4, \dots$	Shell energy level and average distance from nucleus (K, L, M, N shells).
Azimuthal (Angular)	l	$0, 1, 2, \dots, n-1$	Subshell shape: $l = 0$ (s , spherical), $l = 1$ (p , dumbbell), $l = 2$ (d), $l = 3$ (f).
Magnetic	m_l	$-l, \dots, 0, \dots, +l$	Spatial orientation of the orbital in space ($2l + 1$ orbitals per subshell).
Spin	m_s	$+\frac{1}{2}, -\frac{1}{2}$	Intrinsic angular momentum / spin orientation (spin up $\uparrow$ or spin down $\downarrow$).

- **Shell Electron Capacity:**
 - Each individual orbital can accommodate at most **two electrons of opposite spin** (Pauli Exclusion Principle).
 - Subshell capacities: s has 1 orbital (2 electrons); p has 3 orbitals (6 electrons); d has 5 orbitals (10 electrons); f has 7 orbitals (14 electrons).
 - Maximum number of electrons in principal shell n :

$$\text{Capacity} = 2n^2$$

$$(n = 1 \rightarrow 2; \quad n = 2 \rightarrow 8; \quad n = 3 \rightarrow 18; \quad n = 4 \rightarrow 32).$$

C. Electron Configurations & The Periodic Table

Electrons populate energy states according to three foundational physical rules:

1. **Aufbau Principle:** Electrons fill the lowest available quantum energy states first:

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \dots$$

(Note that the $4s$ subshell is lower in energy than $3d$ when unoccupied!)

2. **Pauli Exclusion Principle:** No two electrons in an atom can have the exact same set of all four quantum numbers (n, l, m_l, m_s).
 3. **Hund's Rule:** In subshells containing degenerate orbitals (p, d, f), electrons occupy separate orbitals with parallel spins before pairing up.
- **Valence Electrons:** Electrons occupying the outermost unfilled principal shell. Valence electrons participate in chemical bonding and dictate the electrical, optical, and chemical properties of materials.
 - **Stable Noble Gas Configurations:** Atoms with filled outer s and p subshells (s^2p^6 , 8 valence electrons) exhibit exceptional thermodynamic stability and near-zero chemical reactivity (He, Ne, Ar, Kr, Xe).

D. Electronegativity Trends

Electronegativity (X) is a measure of an atom's ability to attract shared electrons to itself in a chemical bond:

- **Periodic Trend:** Electronegativity **increases from left to right** across a period (increasing nuclear charge pulls electrons tighter) and **decreases from top to bottom** down a group (increased electron shielding weakens nuclear attraction).
- **Extremes:** Fluorine has the highest electronegativity ($X_F = 4.0$ on the Pauling scale), while Francium and Cesium have the lowest ($X \approx 0.7$).

3. Atomic Bonding in Solids: Forces & Potential Energies (Callister §2.3)

3.1 The Interatomic Force & Energy Engine

Consider two isolated atoms brought together from an infinite separation distance ($r = \infty$). The net interaction consists of two opposing forces:

$$F_N(r) = F_A(r) + F_R(r)$$

1. **Attractive Force (F_A):** Operates at long and intermediate distances. Its physical origin depends on the bonding mechanism (electrostatic Coulomb attraction for ionic bonds; electron-nucleus sharing for covalent/metallic bonds).
2. **Repulsive Force (F_R):** Operates at very short interatomic separations. When the outer electron clouds of adjacent atoms overlap, the **Pauli Exclusion Principle** violently resists forcing electrons into already occupied quantum states, creating a sharp electrostatic core repulsion.

Mathematical Potential Energy Relationship:

Potential energy E is related to force F by work integration:

$$E_N(r) = \int_{\infty}^r F_N(r') dr' = E_A(r) + E_R(r)$$

Conversely, force is the negative derivative of potential energy:

$$F_N(r) = -\frac{dE_N(r)}{dr}$$

For many engineering systems, the net potential energy is mathematically modeled by power-law potentials (e.g., Mie or Lennard-Jones potentials):

$$E_N(r) = -\frac{A}{r^m} + \frac{B}{r^n}$$

where $A, B > 0$ are material-specific constants, and $n > m$:

- For simple ionic bonding (Coulomb's Law): $m = 1$.
- The repulsive exponent n typically ranges from 7 to 12 (reflecting the steepness of Pauli core repulsion).

Differentiating yields the net force:

$$F_N(r) = -\frac{d}{dr} \left[-Ar^{-m} + Br^{-n} \right] = - \left[mAr^{-(m+1)} - nBr^{-(n+1)} \right] = -\frac{mA}{r^{m+1}} + \frac{nB}{r^{n+1}}$$

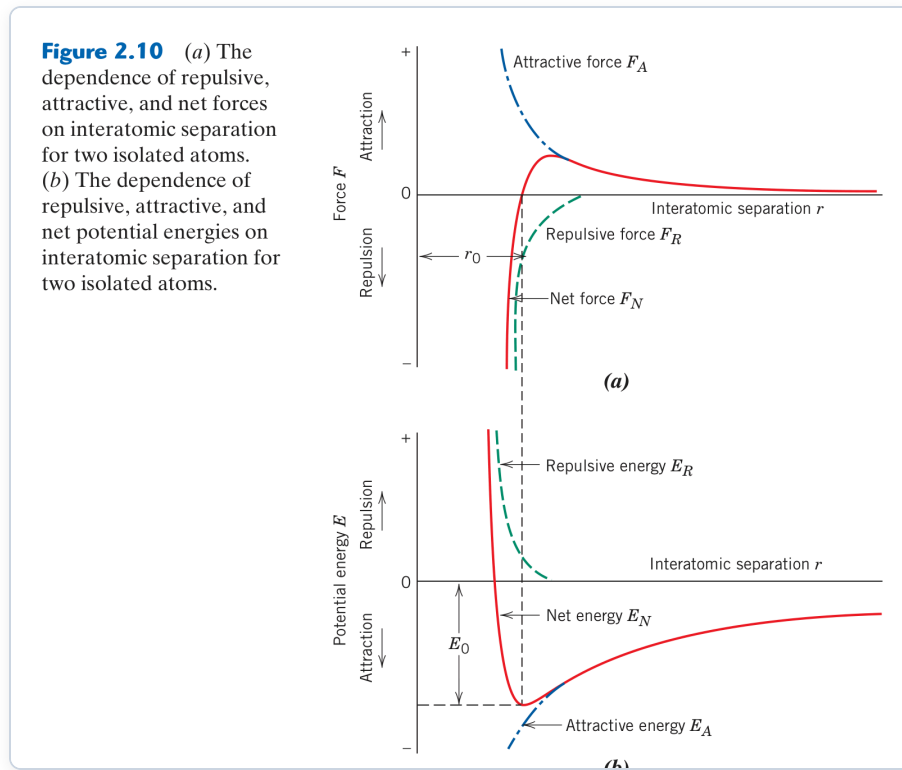


Figure 2.10: (a) Net interatomic force F_N as a function of atomic separation distance r . (b) Net potential energy E_N displaying the equilibrium potential well with bonding energy E_0 and equilibrium spacing r_0 — from Callister & Rethwisch 10th Ed. (Fig. 2.10).

3.2 The Equilibrium State ($r = r_0$)

At the equilibrium interatomic separation $r = r_0$:

1. **Force Balance:** The attractive and repulsive forces exactly balance:

$$F_N(r_0) = 0 \iff F_A(r_0) = -F_R(r_0)$$

2. **Energy Minimum:** The net potential energy reaches its absolute minimum (the bottom of the potential well):

$$\left. \frac{dE_N}{dr} \right|_{r=r_0} = 0$$

Setting the derivative to zero:

$$\frac{mA}{r_0^{m+1}} = \frac{nB}{r_0^{n+1}} \implies r_0^{n-m} = \frac{nB}{mA} \implies r_0 = \left(\frac{nB}{mA} \right)^{\frac{1}{n-m}}$$

3. **Bonding Energy (E_0)**: The depth of the potential energy trough:

$$E_0 = E_N(r_0) = -\frac{A}{r_0^m} + \frac{B}{r_0^n}$$

Substituting $B = \frac{mA}{n} r_0^{n-m}$:

$$E_0 = -\frac{A}{r_0^m} + \frac{mA}{nr_0^m} = -\frac{A}{r_0^m} \left(1 - \frac{m}{n} \right)$$

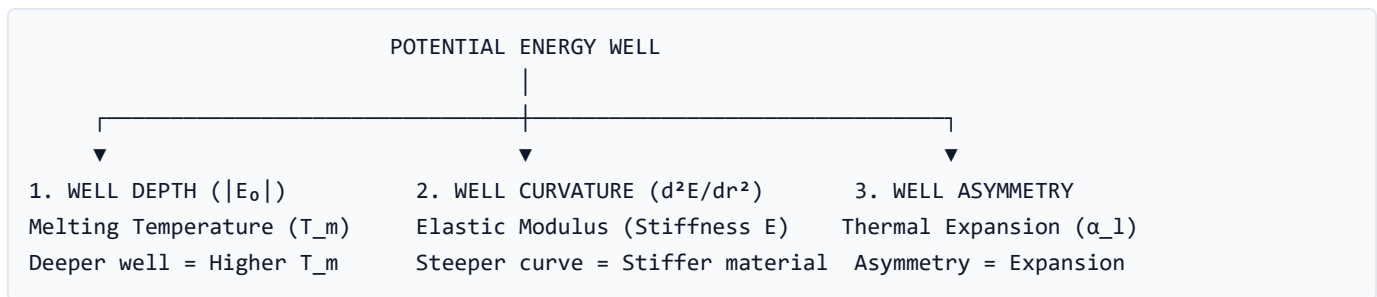
For simple ionic pairs ($m = 1$):

$$E_0 = -\frac{A}{r_0} \left(1 - \frac{1}{n} \right)$$

The magnitude $|E_0|$ is the **bonding energy**—the precise energy required to pull the two bonded atoms infinitely far apart.

3.3 Translating the Potential Energy Well to Macroscopic Properties

The shape of the interatomic potential energy well $E_N(r)$ directly dictates three fundamental macroscopic engineering properties:



1. Melting Temperature (T_m)

- Temperature is a macroscopic measure of the average kinetic energy of atomic vibrations.
- To melt a solid, thermal vibrational energy ($k_B T$) must overcome the cohesive bonding energy holding atoms in their lattice sites.
- **Direct Law:** Materials with deep potential energy wells (large $|E_0|$) exhibit very high melting temperatures:
 - Diamond (Covalent): $|E_0| = 711 \text{ kJ/mol} \implies T_m > 3550^\circ\text{C}$
 - Tungsten (Metallic): $|E_0| = 849 \text{ kJ/mol} \implies T_m = 3410^\circ\text{C}$
 - Silicon Dioxide (Ionic/Covalent): $|E_0| = 464 \text{ kJ/mol} \implies T_m = 1710^\circ\text{C}$
 - Lead (Weak Metallic): $|E_0| = 195 \text{ kJ/mol} \implies T_m = 327^\circ\text{C}$
 - Polyethylene (Secondary vdW between chains): $|E_0| \approx 5 \text{ kJ/mol} \implies T_m \approx 135^\circ\text{C}$

2. Mechanical Stiffness / Elastic Modulus (E)

- When an external tensile force pulls atoms slightly apart by displacement $\Delta r = r - r_0$, the restoring force is governed by Hooke's Law at the atomic scale: $F_N \approx \left(\left(\frac{dF_N}{dr} \right) \right)_{r=r_0} \Delta r = \left(\left(\frac{d^2 E_N}{dr^2} \right) \right)_{r=r_0} \Delta r$

$\left(\frac{d^2 E_N}{dr^2} \right)_{r=r_0} \Delta r$

- Direct Law:** The macroscopic Young's modulus E is directly proportional to the **curvature (second derivative) at the bottom of the potential energy well:**

$$E \propto \left. \frac{d^2 E_N}{dr^2} \right|_{r=r_0}$$

- A narrow, steep potential well acts like an extremely stiff spring, giving high elastic modulus (e.g., Diamond $E \approx 1000$ GPa, Alumina $E \approx 380$ GPa, Steel $E \approx 207$ GPa).
- A wide, shallow well acts like a soft spring, resulting in low modulus (e.g., Polymers $E \approx 1 - 3$ GPa).

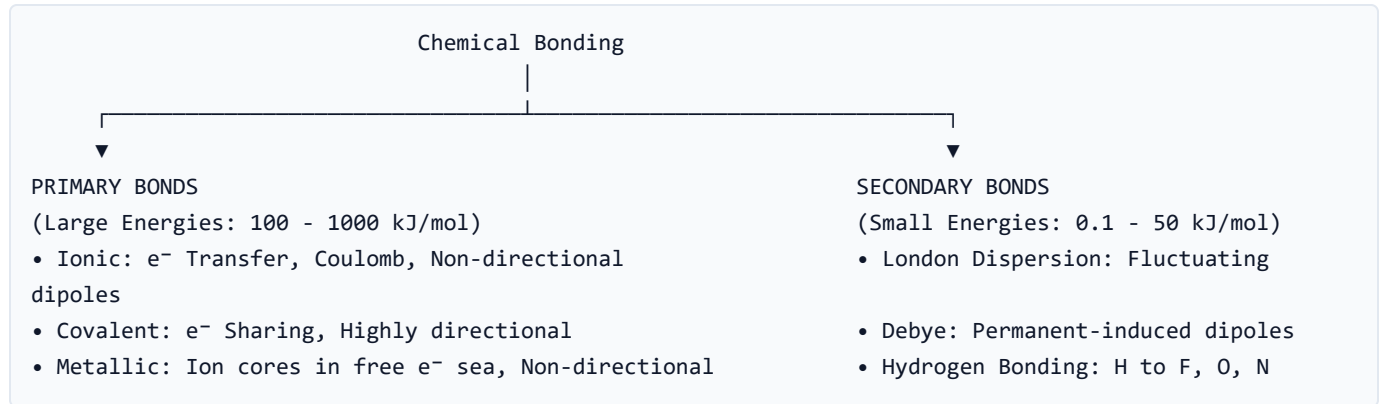
3. Coefficient of Linear Thermal Expansion (α_l)

- In an ideal, symmetric parabolic harmonic well ($E = \frac{1}{2}k(r - r_0)^2$), heating causes atoms to oscillate back and forth with increasing amplitude, but the **mean equilibrium position $\bar{r}$ remains exactly at r_0** . A material with a perfectly symmetric well would exhibit **zero thermal expansion!**
- In real materials, the potential energy well is **asymmetric (anharmonic)**: The repulsive barrier at $r < r_0$ is extremely steep due to Pauli exclusion, while the attractive branch at $r > r_0$ flattens out gradually.
- As thermal energy increases, atoms oscillate with larger amplitude between $r_{\min}$ and $r_{\max}$. Because of the asymmetric flare on the right side:

$$\bar{r}(T) = \frac{r_{\min} + r_{\max}}{2} > r_0$$

- Direct Law:** As temperature rises, the mean interatomic separation $\bar{r}(T)$ shifts outward. The rate of expansion $\alpha_l = \frac{1}{L} \frac{dL}{dT}$ is proportional to the **asymmetry of the well**:
 - Deep, narrow wells are highly symmetric $\implies$ Low thermal expansion (e.g., Invar alloy $\alpha_l \approx 1.2 \times 10^{-6} \text{ K}^{-1}$, Ceramics $\alpha_l \approx 4 - 8 \times 10^{-6} \text{ K}^{-1}$).
 - Shallow, wide wells are highly asymmetric $\implies$ High thermal expansion (e.g., Polymers $\alpha_l \approx 50 - 200 \times 10^{-6} \text{ K}^{-1}$).
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4. The Primary & Secondary Bonding Taxonomy



4.1 Primary Interatomic Bonds

A. Ionic Bonding

- **Mechanism:** Electron transfer from an electropositive metallic element (which loses electrons to form a stable cation, e.g., $\text{Na} \rightarrow \text{Na}^+ + e^-$) to an electronegative nonmetallic element (which gains electrons to form a stable anion, e.g., $\text{Cl} + e^- \rightarrow \text{Cl}^-$).
- **Attractive Coulomb Potential:**

$$E_A(r) = -\frac{|z_1 z_2| e^2}{4\pi\epsilon_0 r} = -\frac{A}{r}$$

where z_1, z_2 are ionic valences, $e = 1.602 \times 10^{-19}$ C, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m. The constant A for monovalent ions ($z_1 = +1, z_2 = -1$) is:

$$A = \frac{(1.602 \times 10^{-19})^2}{4\pi(8.854 \times 10^{-12})} = 2.307 \times 10^{-28} \text{ J} \cdot \text{m} = 1.44 \text{ eV} \cdot \text{nm}$$

- **Characteristics:**
 - Non-directional: Each cation attracts anions equally in all directions in 3D space, governed strictly by geometric packing radius ratios.
 - Predominates in ceramic materials (NaCl , MgO , Al_2O_3 , ZrO_2).
 - Hard and brittle; electrically insulating in solid state because ions are locked in crystal lattice positions.

B. Covalent Bonding

- **Mechanism:** Sharing of valence electrons between adjacent atoms with small electronegativity differences ($X_A \approx X_B$).
- **Characteristics:**
 - **Highly Directional:** Bonds form only along specific spatial angles corresponding to overlapping quantum hybrid orbitals (e.g., sp^3 tetrahedral hybridization in carbon/diamond and silicon with bond angles of 109.5°).
 - High bond strength: Diamond has the highest hardness and highest thermal conductivity of any known bulk material.
 - Low electrical conductivity at room temperature due to tightly bound valence electron pairs.

C. Pauling's Percent Ionic Character Formula

Real chemical bonds between dissimilar atoms are rarely 100% purely ionic or 100% purely covalent. The degree of ionic character depends exponentially on the **difference in Pauling electronegativities** ($\Delta X = |X_A - X_B|$):

- If $\Delta X = 0$ (e.g., C-C, Si-Si): (Pure Covalent).
- If $\Delta X \approx 1.7$: (Half ionic, half covalent).
- If $\Delta X > 2.0$ (e.g., NaCl with $X_{\text{Cl}} = 3.0$, $X_{\text{Na}} = 0.9 \implies \Delta X = 2.1$): (Predominantly ionic).

D. Metallic Bonding

- **Mechanism:** Valence electrons are shed by metallic atoms to form positive ion cores; the freed electrons delocalize into a shared, highly mobile **conduction electron gas (electron sea)**.
- **Characteristics:**
 - Non-directional: Ion cores are bound by electrostatic attraction to the pervasive negative electron gas.
 - Exceptional ductility and formability: When a shear stress is applied, planes of positive ion cores slide past one another smoothly because the flexible electron sea instantly readjusts without bond breakage.
 - High electrical and thermal conductivities: Free electrons move freely through the lattice under electric fields or temperature gradients.

4.2 Secondary (van der Waals) Bonding

Secondary bonds are physical bonds arising from **electrostatic attraction between atomic or molecular dipoles**. They are an order of magnitude weaker than primary chemical bonds (0.1 – 50 kJ/mol vs. 100 – 1000 kJ/mol):

1. Fluctuating Induced Dipoles (London Dispersion Forces):

- Instantaneous fluctuations in electron cloud symmetry generate temporary, short-lived dipoles that induce complementary dipoles in adjacent atoms.
- Weakest bond; present in all matter, but the only bond holding liquid noble gases (Argon, Helium) and nonpolar polymer chains together.

2. Permanent Dipole Bonds & Hydrogen Bonding:

- Polar molecules (e.g., HCl, H₂O) possess asymmetric charge distributions creating permanent electric dipoles.
 - **Hydrogen Bonding:** A special, exceptionally strong category of permanent dipole bonding occurring when hydrogen is covalently bonded to highly electronegative, small atoms (**Fluorine, Oxygen, Nitrogen**). The unshielded proton forms an intense localized positive charge that strongly attracts the negative lone-pair electrons of adjacent molecules.
 - Hydrogen bonding explains why water is liquid at room temperature ($T_b = 100^\circ\text{C}$) while heavier hydride analogs (H₂S) are gases ($T_b = -60^\circ\text{C}$).
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5. Comprehensive Step-by-Step Problem Walkthroughs

5.1 Problem 1: Quantitative Interatomic Force & Energy Calculation

Problem Statement: The net potential energy $E_N(r)$ between two isolated monovalent ions (K^+ and Cl^-) is modeled by:

$$E_N(r) = -\frac{C}{r} + \frac{D}{r^8}$$

where r is the interionic separation distance in nanometers (nm), attractive constant $C = 1.436 \text{ eV} \cdot \text{nm}$, and repulsive constant $D = 7.32 \times 10^{-6} \text{ eV} \cdot \text{nm}^8$.

1. Derive the expression for the net interatomic force $F_N(r)$.
2. Calculate the equilibrium separation distance r_0 (in nm).
3. Calculate the bonding energy E_0 (in eV).
4. Calculate the net force at $r = 0.20 \text{ nm}$ and state whether it is attractive or repulsive.

Step 1: Derive the Net Interatomic Force $F_N(r)$

Force is the negative gradient of potential energy:

$$F_N(r) = -\frac{dE_N}{dr} = -\frac{d}{dr} [-Cr^{-1} + Dr^{-8}]$$

$$F_N(r) = -[Cr^{-2} - 8Dr^{-9}] = -\frac{C}{r^2} + \frac{8D}{r^9}$$

- Attractive force: $F_A(r) = -\frac{C}{r^2} = -\frac{1.436}{r^2} \text{ eV/nm}$
- Repulsive force: $F_R(r) = +\frac{8D}{r^9} = +\frac{8(7.32 \times 10^{-6})}{r^9} = +\frac{5.856 \times 10^{-5}}{r^9} \text{ eV/nm}$

Step 2: Calculate Equilibrium Separation Distance r_0

At equilibrium $r = r_0$, the net force is zero ($F_N(r_0) = 0$):

$$-\frac{1.436}{r_0^2} + \frac{5.856 \times 10^{-5}}{r_0^9} = 0$$

Multiply through by r_0^9 :

$$-1.436r_0^7 + 5.856 \times 10^{-5} = 0$$

$$r_0^7 = \frac{5.856 \times 10^{-5}}{1.436} = 4.078 \times 10^{-5} \text{ nm}^7$$

Take the 7th root of both sides:

$$r_0 = (4.078 \times 10^{-5})^{1/7} = 0.2364 \text{ nm} = 2.364 \text{ \AA}$$

Step 3: Calculate the Bonding Energy $E_0 = E_N(r_0)$

Substitute $r_0 = 0.2364$ nm into the potential energy equation:

$$E_0 = -\frac{1.436}{0.2364} + \frac{7.32 \times 10^{-6}}{(0.2364)^8}$$

- Attractive potential: $E_A = -\frac{1.436}{0.2364} = -6.074$ eV
- Repulsive potential:

$$(0.2364)^8 = 9.641 \times 10^{-6} \text{ nm}^8$$

$$E_R = \frac{7.32 \times 10^{-6}}{9.641 \times 10^{-6}} = +0.759 \text{ eV}$$

- Net bonding energy:

$$E_0 = -6.074 \text{ eV} + 0.759 \text{ eV} = -5.315 \text{ eV}$$

The depth of the potential energy well is $|E_0| = 5.32$ eV (513 kJ/mol).

Step 4: Calculate Net Force at $r = 0.20$ nm

Since $r = 0.20$ nm $<$ $r_0 = 0.2364$ nm, the ions are squeezed closer than equilibrium; we expect strong repulsion:

$$F_N(0.20) = -\frac{1.436}{(0.20)^2} + \frac{5.856 \times 10^{-5}}{(0.20)^9}$$

- $F_A = -\frac{1.436}{0.04} = -35.90$ eV/nm
- $(0.20)^9 = 5.12 \times 10^{-7} \text{ nm}^9$
- $F_R = \frac{5.856 \times 10^{-5}}{5.12 \times 10^{-7}} = +114.38$ eV/nm

$$F_N = -35.90 + 114.38 = +78.48 \text{ eV/nm}$$

Convert to SI Newtons ($1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$, $1 \text{ nm} = 10^{-9} \text{ m}$):

$$F_N = 78.48 \times \frac{1.602 \times 10^{-19} \text{ J}}{10^{-9} \text{ m}} = 1.26 \times 10^{-8} \text{ N} = 12.6 \text{ nN}$$

Because $F_N > 0$, the net force is **strongly repulsive**, acting to push the two ions back out to r_0 .

5.2 Problem 2: Pauling's Percent Ionic Character for Engineering Ceramics

Problem Statement: Using the Pauling electronegativity values provided in the table below, compute the percent ionic character (%) for the following engineering ceramics:

1. Silicon Carbide (SiC)
2. Aluminum Oxide (Al_2O_3)
3. Magnesium Oxide (MgO)

4. Gallium Arsenide (GaAs)

Element	Electronegativity (X)	Element	Electronegativity (X)
Silicon (Si)	1.8	Oxygen (O)	3.5
Carbon (C)	2.5	Magnesium (Mg)	1.2
Aluminum (Al)	1.5	Gallium (Ga)	1.6
Arsenic (As)	2.0	Chlorine (Cl)	3.0

Step-by-Step Computations via Pauling's Formula:

1. Silicon Carbide (SiC):

$$\Delta X = |X_C - X_{Si}| = |2.5 - 1.8| = 0.7$$

$$(\Delta X)^2 = 0.49$$

Classification

: Highly covalent structural ceramic with extreme hardness.

2. Aluminum Oxide (Al₂O₃):

$$\Delta X = |X_O - X_{Al}| = |3.5 - 1.5| = 2.0$$

$$(\Delta X)^2 = 4.0$$

Classification

: Predominantly ionic ceramic insulator.

3. Magnesium Oxide (MgO):

$$\Delta X = |X_O - X_{Mg}| = |3.5 - 1.2| = 2.3$$

$$(\Delta X)^2 = 5.29$$

Classification

: Strongly ionic refractory ceramic.

4. Gallium Arsenide (GaAs):

$$\Delta X = |X_{As} - X_{Ga}| = |2.0 - 1.6| = 0.4$$

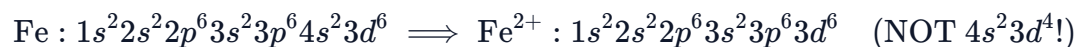
$$(\Delta X)^2 = 0.16$$

Classification

: Almost purely covalent compound semiconductor.

6. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Transition Metal Ionization Order:** When determining electron configurations for transition metal cations (e.g., Fe^{2+} or Cu^+), **electrons are always removed from the outermost valence shell (4s) before the underlying (3d) shell:**



- **⚠ Trap 2: Force Derivative Sign Confusion:** Remember that $F_N = -\frac{dE_N}{dr}$. The negative sign is crucial:
 - When $r > r_0$, the slope $\frac{dE_N}{dr}$ is positive $\implies F_N < 0$ (attractive force).
 - When $r < r_0$, the slope $\frac{dE_N}{dr}$ is negative $\implies F_N > 0$ (repulsive force).
 - If you omit the negative sign, you will report repulsive forces as attractive and vice versa!
- **⚠ Trap 3: Equating Bonding Energy to Melting Point Directly Without Well Depth:** Melting point scales with **well depth** $|E_0|$, NOT the separation distance r_0 . Two materials can have the same r_0 , but if material A has twice the well depth of material B, material A will have a drastically higher melting temperature.
- **⚠ Trap 4: Missing Units in Coulombic Constant A :** In $E_A = -\frac{A}{r}$, ensure that if r is in **nm**, A must be in $\text{eV} \cdot \text{nm}$ ($A \approx 1.44 \text{ eV} \cdot \text{nm}$). If r is in meters (m), A must be in Joules-meters ($A = \frac{e^2}{4\pi\epsilon_0} = 2.307 \times 10^{-28} \text{ J} \cdot \text{m}$). Mixing nm with Joules produces errors of 9 orders of magnitude!